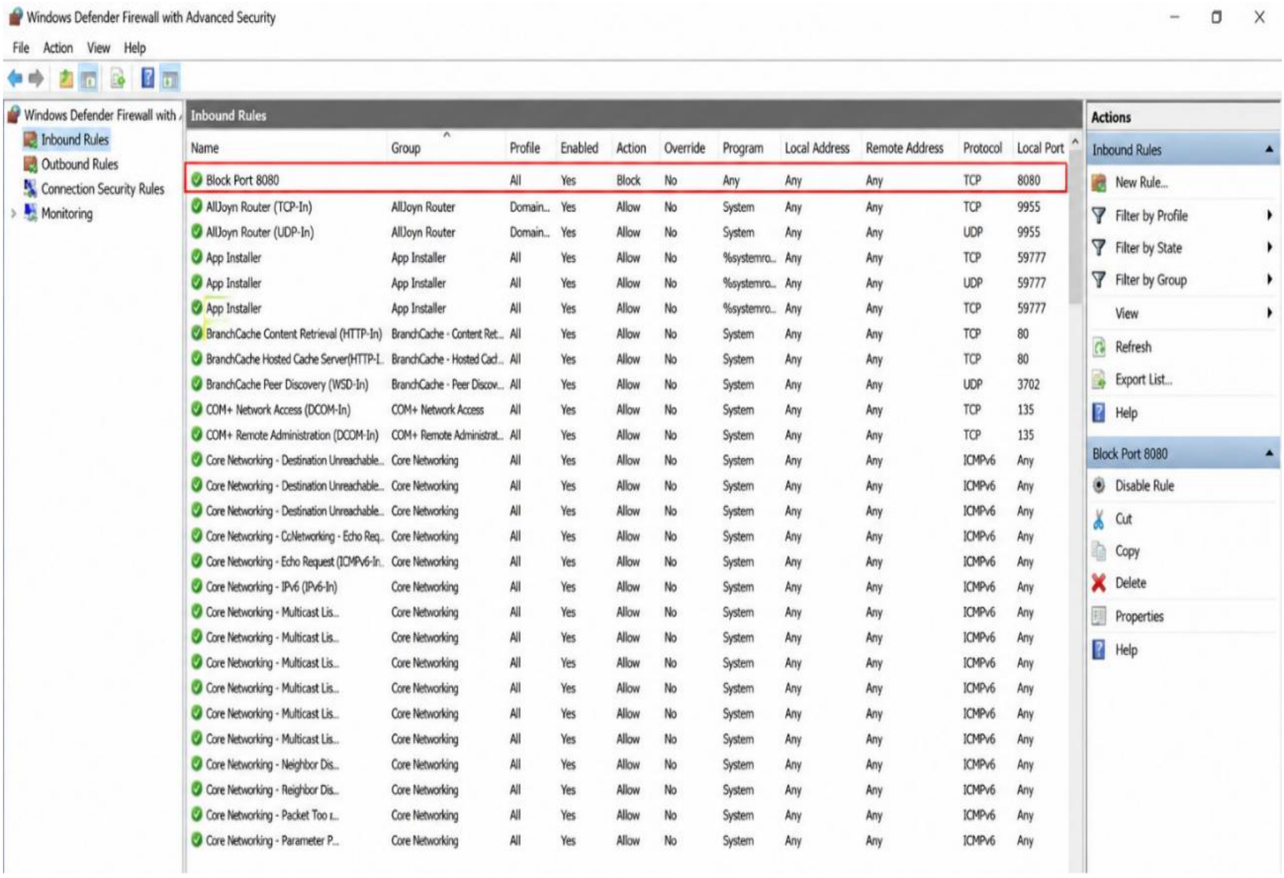
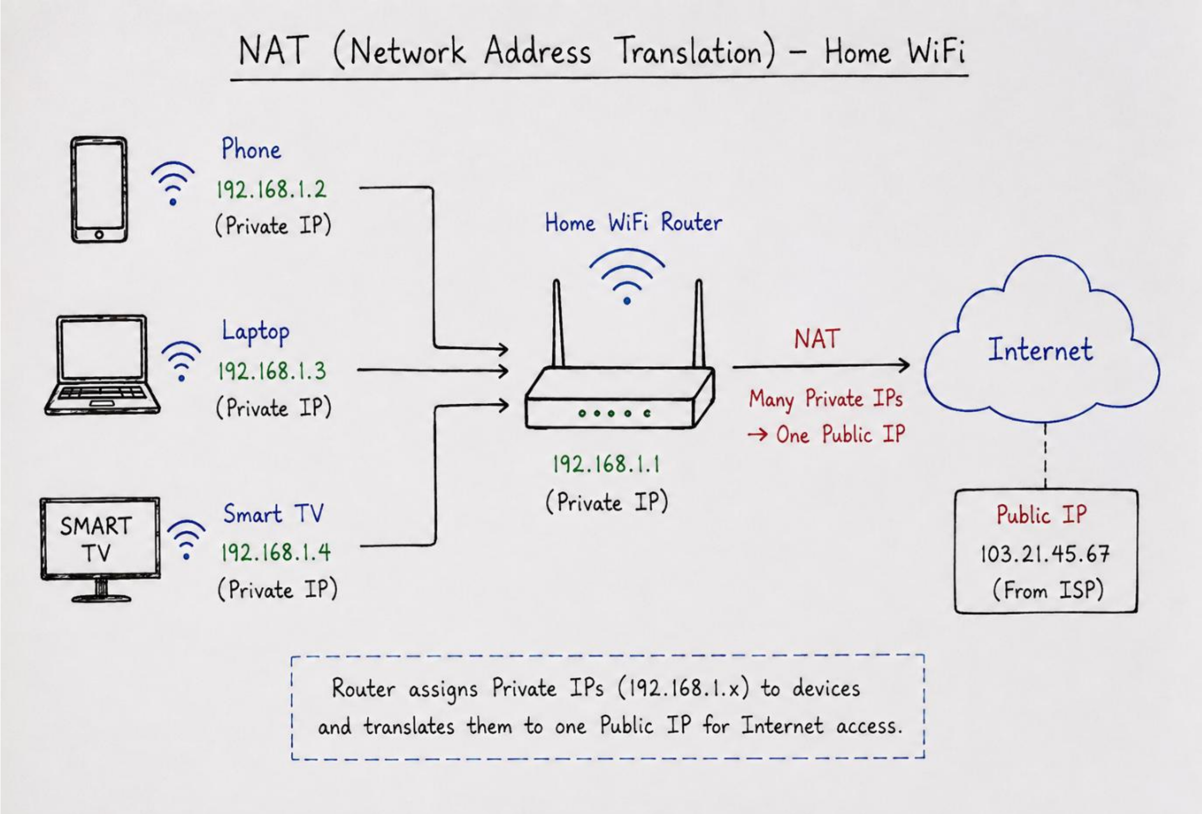
Task

1) Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2) Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.



3) Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.
   - Log in to router's admin panel by opening a browser and entering the router's IP address (for example, 192.168.1.1).
   - Enter the administrator username and password.
   - Go to Security, Firewall, Access Control, or Port Filtering.
   - Select Port Filtering or Outbound Rules.
   - Click Add New Rule.

- Configure the rule as follows:
- Rule Name: Block SMTP
- Direction: Outbound
- Protocol: TCP
- Port: 25
- Action: Deny / Block
- Save the rule and apply the changes.
- Restart the router if required.

Setting	Value
Rule Name	Block SMTP
Direction	Outbound
Protocol	TCP
Port	25
Action	Block

4) Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.

Note : We can't determine which packets are **allowed** or **blocked** without the **packet filtering rule set**. The decision depends entirely on the firewall rules (source IP, destination IP, protocol, ports, and action).

for example:

Rule	Source IP	Destination IP	Protocol	Port	Action
1	Any	Any	TCP	80	Allow
2	Any	Any	TCP	22	Block
3	Any	Any	UDP	53	Allow

These three packets:

1. **HTTP request** from **192.168.1.5 → 172.217.166.78:80**
2. **SSH request** from **10.0.0.10 → 34.201.23.1:22**
3. **DNS query** from **192.168.1.7 → 8.8.8.8:53**

5) Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

⇒ Here are three common ways **Instagram** and **Zomato** may use **firewalls** and **packet filtering** to protect their servers and users.

❖ Blocking Unauthorized Access

❖ Firewalls allow only trusted network traffic to reach the servers. If someone tries to connect using unauthorized ports or suspicious IP addresses, the firewall blocks the request before it reaches the application.

❖ Preventing DDoS (Distributed Denial-of-Service) Attacks

❖ Both Instagram and Zomato can use firewalls and packet filtering to detect unusually high volumes of traffic from suspicious sources. Malicious traffic is filtered or rate-limited, helping keep the website and mobile app available for legitimate users.

❖ Filtering Malicious Traffic

❖ Packet filtering checks incoming and outgoing network packets based on rules such as IP address, port number, and protocol. Traffic that appears to be malicious—such as scanning attempts, fake requests, or packets from blacklisted IP addresses—is blocked to reduce the risk of attacks and misuse.